

Signed quitting: normalisation boundary and explicit symmetric-prior counterexamples

Question 26 of *Intelligence via Ultrafilters* under the reward-punishment-symmetric normalisation

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Provenance. Produced by OpenAI Codex during an external audit of this repository (27 September 2026), and checked by Anthropic Claude before inclusion. Hand proofs; no independent human review; priority not established.

Result. For every real r there exist a symmetric universal reference machine U and computable agents A and B violating Question 26’s implication under the expected-value normalisation. This proves Conjecture 5.4 of note 09 in that quantifier form. It is not a proof for every pre-assigned U , nor for one U working for all r . Under Legg and Hutter’s pathwise normalisation (note 01), the implication holds for $r \geq 0$.

Sources: [Intelligence via ultrafilters](#), Definitions 17–19 and Question 26; [Alexander-Hutter, Reward-punishment symmetric universal intelligence](#), Definitions 1–2 and 10–12. This combines the first paper’s designated skip action with the second paper’s **expected-value** normalization. Assume at least two actions and two observations, and reward alphabet $\{-1, 0, 1\}$.

1. Which normalization?

The 2021 paper requires, for every agent π , $V_\mu^\pi = \lim_n \mathbb{E} S_n \in [-1, 1]$. It does not require every realized trajectory to total at most 1.

For a deterministic skip-respecting environment with totals bounded by 1 for every agent, every reachable partial sum is also at most 1: follow its action prefix and then skip forever. The same argument applies under a genuinely pathwise total bound in a stochastic environment. Consequently, in that stronger model:

- If $r > 0$, A_{r+1} never quits on a reachable history.
- If $r = 0$, A_1 can quit only at total 1, and doing so cannot lower its eventual value, even when later rewards can be negative.

Thus Question 26’s implication holds for $r \geq 0$ in that model, regardless of the condition on B_r . The proposed universal failure is impossible there.

Under the actual **expected-value** normalization, large partial sums can occur on rare branches. For example, with probability $1/L$ an environment offers L unit rewards, otherwise zero. Every agent’s expected value is at most 1, but the successful branch can total L . This is the resource used below.

2. Symmetry facts and the universal tail

Let W_{skip} be the computable well-behaved environments respecting skipping. It is closed under reward duality $\mu \mapsto \bar{\mu}$. Fix a symmetric universal machine U_0 for the paper's suitable RL encoding, and write

$$\Gamma'_U(\pi) = \sum_{\mu \in W_{skip}} 2^{-K_U(\mu)} V_{\mu}^{\pi}.$$

The series converges absolutely. For a policy Q invariant under negating all past rewards, dual environments give opposite values, so $\Gamma'_U(Q)=0$ whenever U is symmetric. In particular Skip has value zero.

For a chosen non-self-dual pair μ_+ , $\mu_- = \bar{\mu}_+$, give them descriptions 00 and 01 . Let the only other branch be $1^m p$, simulating $U_0(p)$, with $m \geq 3$. This gives a prefix-free universal machine U . Both special complexities are exactly 2. Every other environment and its dual have complexities $m + K_{U_0}(\mu)$, hence U remains symmetric. The total weight of all nonspecial environments is at most 2^{-m} , by Kraft's inequality. Since their values are in $[-1, 1]$, their combined contribution to any agent lies in $[-2^{-m}, 2^{-m}]$. This is the signed version of note 02's selector-tail construction; it does not require recognizing shortest programs.

3. Counterexamples for $r \leq 0$

Set $c=r+1 \leq 1$ and $B=Skip$. Let A take action 1 iff its **initial reward equals 1**, and otherwise always skip. A is computable and does not need to compute r .

On every skip-respecting environment, A_c behaves as Skip: when the initial reward is 1 it already meets the threshold, and otherwise A itself skips. Thus $\Gamma'_U(A_c) = \Gamma'_U(B) = 0$, and $B_r = B$.

The special μ_+ tosses a fair coin initially. On success it gives initial reward 1 and a success observation, then offers one further unit reward for action 1 on the first turn; on failure it gives zero and never rewards. It then gives zero forever. Skips receive zero; any agent has expected value in $[0, 1]$. Its reward dual μ_- is also well-behaved.

For A , the values are 1 in μ_+ and $-1/2$ in μ_- : it refuses the extra punishment after the negative initial reward. The special weighted contribution is $1/8$. Choose $m=4$. Then $\Gamma'_U(A) \geq 1/8 - 1/16 > 0$, proving the required failure.

The $r=0$ case specifically separates this signed expected-value model from the pathwise Legg-Hutter convention in note 01.

4. Counterexamples for $r > 0$

Put $c=r+1 > 1$, $K=\lfloor c \rfloor$ and $L=K+1$. All rewards and partial sums are integers. Define the computable agent A to act 1 until a partial sum has reached $-K$, then skip forever. Let Q act 1 until the **absolute value** of a partial sum has reached K , then skip forever. Q is reward-dual invariant.

On every skip-respecting environment, A_c has the same trajectory as Q : positive stopping occurs at integer threshold K , and negative stopping is already built into A . After either event skipping freezes the reward sum. Therefore $\Gamma'_U(A_c) = \Gamma'_U(Q) = 0$ for every symmetric U . As before, take $B=Skip$, so $B_r = B$ and $\Gamma'_U(B)=0$.

The special μ_+ outputs initial reward zero and a success flag with probability $p=1/L$. On success it offers reward +1 for action 1 on each of the next L turns, and zero for skips; on failure all rewards are zero. Thereafter all rewards are zero. Against every agent its expected value lies in $[0, pL]=[0,1]$. Its dual μ_- offers the corresponding negative rewards and has all values in $[-1,0]$. Both are computable, use the fixed finite alphabets, and are well-behaved against stochastic agents as well.

Conditional on success:

Policy	μ_+ total	μ_- total
A	L	-K
A_C , equivalently Q on reachable histories	K	-K
Skip	0	0

The special contribution to $\Gamma'_U(A)$ is $\frac{1}{4}p(L - K) = 1/(4L)$. Choose $m \geq 3$ with $2^{-m} < 1/(4L)$. Then

$$\Gamma'_U(A) \geq \frac{1}{4L} - 2^{-m} > 0 = \Gamma'_U(B) = \Gamma'_U(A_{r+1}), \quad \Gamma'_U(B_r) = \Gamma'_U(B).$$

This proves the counterexample for each $r > 0$. The finite integer K can be built into A even if r is not a computable real; no uniform algorithm from arbitrary real descriptions of r is claimed. ■

What is and is not resolved

Under pathwise bounded totals, failure for every r is false. Under the 2021 expected-value normalization, it is realizable for every r with an explicitly chosen symmetric universal prior. The claim for **each fixed arbitrary** symmetric prior remains open: uncontrolled tail contributions could cancel or reverse A 's advantage. The constructions do not settle Questions 27-28 or sign computability for all symmetric machines.

Exact finite controls (`../audit-2026-09-27/controls.py`) check the special-pair arithmetic, threshold cases and universal-tail margins. Symmetric cancellation over all environments is the hand argument above, not an enumeration.